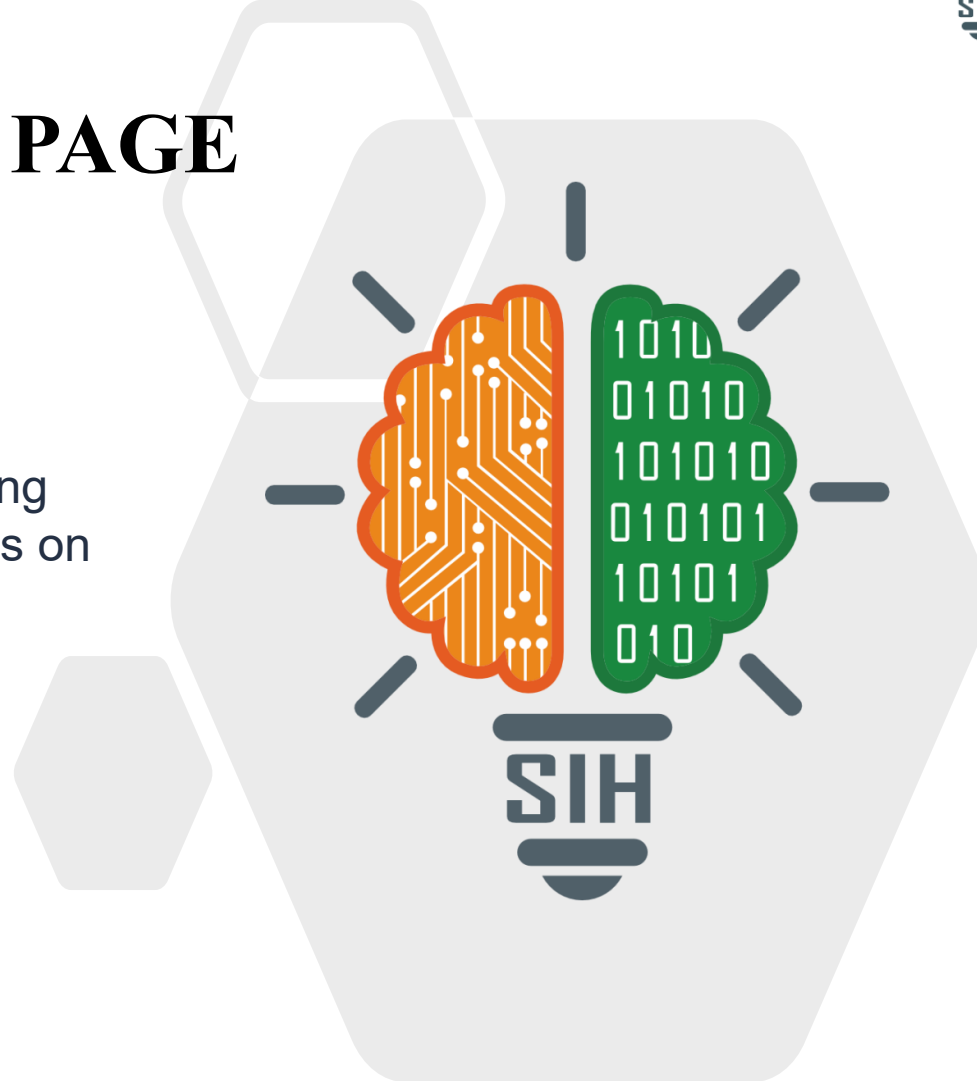


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26037
- **Problem Statement Title**- Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme**- Smart Vehicles
- **PS Category**- Software
- **Team ID**-
- **Team Name** (Registered on portal)



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Threat-aware perception and path planning for mixed Indian traffic — it reacts to conflicts with the ego path, not to mere proximity.

Detailed explanation of the proposed solution

- Detects and tracks every road user from one front camera — cars, autos, two-wheelers, pedestrians, cattle
- Estimates distance, closing speed and time-to-collision; builds a bird's-eye risk map
- Scores 17 candidate paths every frame → GO / SLOW DOWN / BRAKE / NO SAFE PATH, with the reason

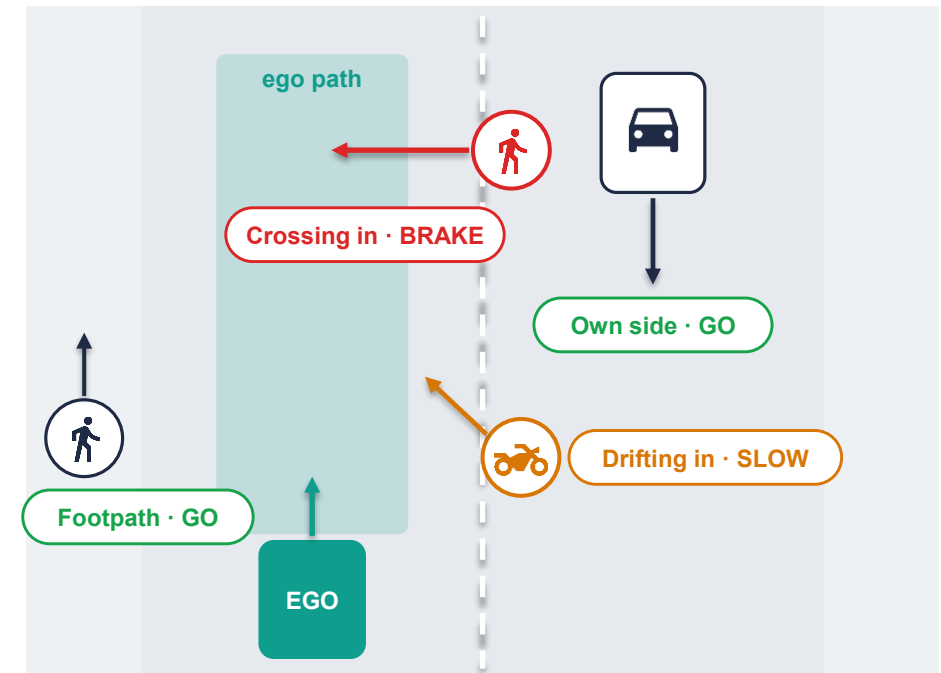
How it addresses the problem

- No lane markings needed: free-space corridor + candidate trajectories
- Irregular motion read from track history: cut-in, crossing, oncoming, wrong-way
- Left-hand-traffic aware: oncoming vehicles on their side and footpath pedestrians do not trigger braking

Innovation and uniqueness of the solution

- Path-threat classes: IN PATH · ENTERING · APPROACHING · NEAR · CLEAR
- Explainable, replayable decisions + a safety regression suite
- Confirmation-first accident response (simulation) and phone-camera live mode

Threat to the ego path — not proximity

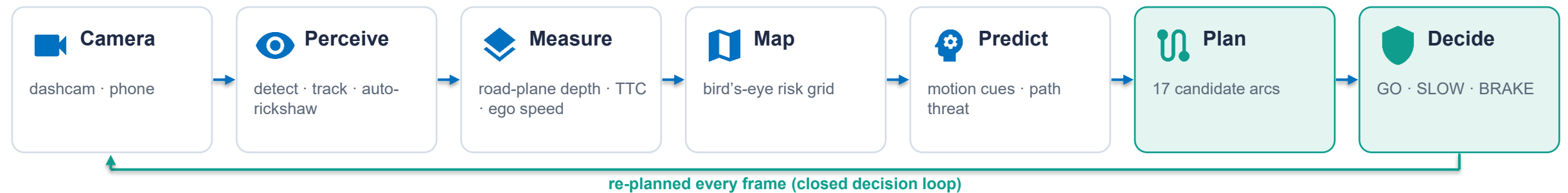


India drives on the left: oncoming traffic is expected on the right. Only a predicted conflict with the ego corridor changes the decision.

Technologies to be used (e.g. programming languages, frameworks, hardware)

Languages	AI models (pretrained, open)	Frameworks	Hardware	Next (PS-recommended)
Python 3.10	YOLOv8n + ByteTrack	PyTorch (CUDA)	1 front camera / phone	MATLAB / Simulink
JavaScript	Depth-Anything-V2	OpenCV	Laptop GPU (RTX 3050)	RoadRunner
	CLIP ViT-B/32	Flask web app		Automated Driving Toolbox

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — measured

75/75	8.6–9.0	11.1 FPS · 73 ms	2.4–2.6 ms	6
safety regression scenarios pass	FPS offline, 720p, laptop GPU	live phone→PC mode (same-PC benchmark)	re-planning time per frame	real drives validated (CC-licensed)

Analysis of the feasibility of the idea



Technically proven

End-to-end prototype runs today: video or phone camera in, explained decision out.



Low cost

One camera and a commodity GPU; open pretrained models — no custom training needed.



Modular

Each stage (detect, depth, plan, decide) is swappable — ready for sensor fusion or MATLAB.

Potential challenges and risks

- Monocular depth is an estimate, not a measurement
- Not yet real time on a laptop GPU
- Unusual road users (pushcarts, animals)
- Over- or under-braking in dense traffic
- Validation limited to recorded video
- False accident alarms

Strategies for overcoming these challenges

- ● Road-plane calibration today; radar/LiDAR fusion next
- ● Model export (ONNX/TensorRT), lower input size, embedded GPU
- ● Fine-tune on the Indian Driving Dataset (IDD)
- ● Path-threat model + 75-scenario regression suite on every change
- ● Closed-loop RoadRunner / Simulink scenes for all five PS scenarios
- ● Multi-cue confirmation + human confirmation; simulation-only today



Potential impact on the target audience



Pedestrians & two-wheelers

Conflicts with the path are flagged early; people beside the road are not.



Drivers & fleets

Taxis, autos, logistics: driver-assist warnings that explain why.



ADAS / AV developers

Indian-road decision logic and a reusable scenario test suite.



Emergency response

Confirmation-first accident alerts with location and nearby hospitals.

Why it matters — India, 2022

4,61,312

road accidents

1,68,491

people killed (MoRTH, Road Accidents in India 2022)

Benefits of the solution (social, economic, environmental, etc.)



Social

- Safer mixed traffic for vulnerable road users
- Decisions come with reasons — builds trust and aids audits



Economic

- Camera-first, open-source stack on commodity hardware
- Phone camera works for demos and training



Environmental

- Fewer unnecessary hard brakes → smoother driving (expected, not yet measured)
- Software upgrade path — no new hardware for pilots

Details / Links of the reference and research work

Perception & models

- Jocher et al., Ultralytics YOLOv8 — github.com/ultralytics/ultralytics
- Zhang et al., ByteTrack, ECCV 2022 — [arXiv:2110.06864](https://arxiv.org/abs/2110.06864)
- Yang et al., Depth Anything V2, NeurIPS 2024 — [arXiv:2406.09414](https://arxiv.org/abs/2406.09414)
- Radford et al., CLIP, ICML 2021 — [arXiv:2103.00020](https://arxiv.org/abs/2103.00020)

Planning & decision

- Paden et al., Motion planning & control for self-driving urban vehicles, IEEE T-IV 2016 — [arXiv:1604.07446](https://arxiv.org/abs/1604.07446)
- Kinematic bicycle model for candidate-arc generation (17 arcs, up to 15 degrees steering)

Data, tools & context

- SIH 2026 PS 26037 (MathWorks) — problem statement & expected solution
- Varma et al., Indian Driving Dataset (IDD), WACV 2019 — idd.insaan.iit.ac.in
- MathWorks RoadRunner & Automated Driving Toolbox — mathworks.com (planned)
- MoRTH, Road Accidents in India 2022 — morth.nic.in
- OpenStreetMap Overpass API — nearby-hospital lookup

Validation footage (Wikimedia Commons)

- Bengaluru & Karnataka drives — L. Shyamal (CC0); Delhi cattle — Fowler&fowler (CC BY-SA 3.0)
- Frederiksted pier (left-hand traffic) — John Edwards (CC BY 3.0), 45 s excerpt

Prototype status: research prototype — not a certified autonomous-driving or safety system; emergency features run in simulation only.